

Where Sorting Creates Value in Production–Waste Flow Networks

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With AI assistance

Abstract—This note starts from an abstract mathematical model of a class of flow networks with sources, processes, consumers, sorting/disassembly operations, and sinks. It then connects that model to reported data from real economies. The target question is the value of sorting and disassembly at different points in the network. The paper does not propose a new model class; it synthesizes established reverse-logistics and closed-loop supply-chain frameworks with material-flow accounting and current waste statistics. The key output is a decision metric: the marginal system value of enabling sorting or deeper disassembly at a given node. Evidence from global e-waste, EU treatment/recycling statistics, and U.S. municipal and construction-demolition streams shows a consistent pattern: value concentrates at nodes where separation prevents irreversible mixing and where downstream recovery markets are already viable.

Index Terms—closed-loop supply chain, reverse logistics, disassembly, sorting, material flow analysis, waste management

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I. ABSTRACT MODEL OF THE NETWORK CLASS

Let

$$G = (V, E), \quad (1)$$

$$V = V^{\text{src}} \cup V^{\text{mfg}} \cup V^{\text{use}} \cup V^{\text{col}} \cup V^{\text{sort}} \cup V^{\text{proc}} \cup V^{\text{sink}}. \quad (2)$$

be a directed multi-layer graph. Each flow has material type $m \in M$, quality state $q \in Q$, and time index $t \in T$. Flow on edge (i, j) is

$$x_{ij}^{m,q,t} \geq 0. \quad (3)$$

Node-level conservation is

$$\sum_{i:(i,v) \in E} x_{iv}^{m,q,t} + u_v^{m,q,t} = \sum_{j:(v,j) \in E} x_{vj}^{m,q,t} + \ell_v^{m,q,t}, \quad (4)$$

where u is external injection (virgin feedstock, imports, returns) and ℓ is unrecoverable loss.

This structure matches established reverse-logistics and closed-loop modeling traditions [1], [2], [3], [4], [5].

II. PROCESS OPERATORS, SORTING, AND DISASSEMBLY DEPTH

At each process node v , transformation from incoming to outgoing material-quality vectors is

$$\mathbf{x}_{v,t}^{\text{out}} = A_v(d_v) \mathbf{x}_{v,t}^{\text{in}}, \quad (5)$$

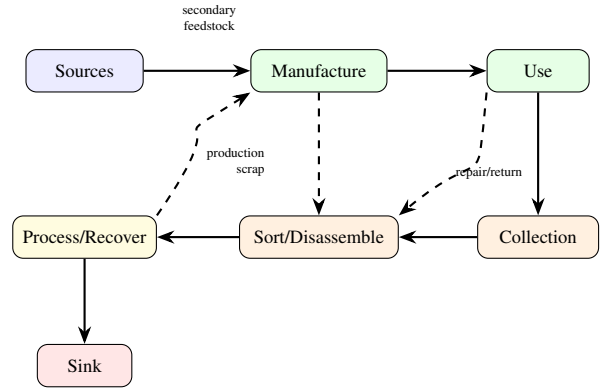


Fig. 1. Abstract flow-network class used in the paper.

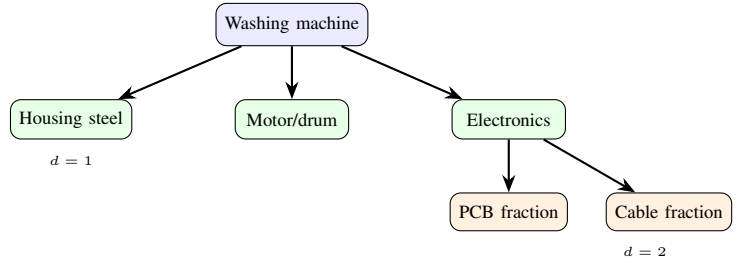


Fig. 2. Illustrative disassembly-depth ladder encoded by $A_v(d_v)$.

where $d_v \in \{0, 1, 2, \dots\}$ is disassembly/sorting depth and $A_v(d_v)$ captures yields, contamination, and quality split.

In this formulation:

- **sorting** changes class labels with limited physical decomposition,
- **disassembly** physically separates modules/materials,
- larger d_v usually improves quality purity but raises handling cost.

III. OBJECTIVE AND VALUE OF SORTING

Define a net system objective (economic with optional externality terms):

$$Z = \sum_t \left[\sum_{(i,j),m,q} p_j^{m,q,t} x_{ij}^{m,q,t} + \sum_{v,m} d_v^{m,t} r_v^{m,t} \right] \quad (6)$$

$$- \sum_{(i,j),m,q} c_{ij}^{m,q,t} x_{ij}^{m,q,t} - \sum_v k_v(d_v, t) - \sum_{v,m} h_v^{m,t} \ell_v^{m,t} \quad (7)$$

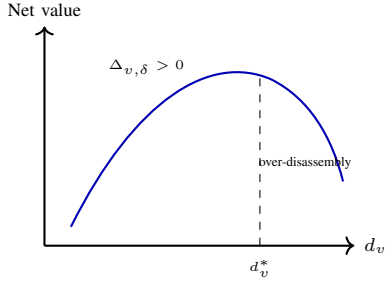


Fig. 3. Stylized depth-value relationship: optimal disassembly depth is finite.

For candidate sorting node v :

$$\Delta_v = Z^*|_{d_v > 0} - Z^*|_{d_v = 0}, \quad (8)$$

and for incremental depth:

$$\Delta_{v,\delta} = Z^*(d_v = \delta) - Z^*(d_v = \delta - 1). \quad (9)$$

Δ_v is the practical ranking metric: where adding sorting capability gives the largest positive marginal effect.

IV. CONNECTING THE MODEL TO REAL ECONOMIES

The model maps to measurable data as follows:

- 1) $x_{ij}^{m,q,t}$ from waste and material-flow statistics.
- 2) $A_v(d_v)$ from observed recovery yields and purity rates.
- 3) c, k from collection, transport, handling, and processing costs.
- 4) p from secondary-material and reusable-component revenues.
- 5) a, h from avoided disposal/virgin production impacts and policy costs.

Policy constraints (collection targets, treatment definitions, EPR obligations) act as model boundary conditions, not side notes [6].

V. EMPIRICAL ANCHORS

A. Global product-to-waste dynamics

World Bank estimates global municipal solid waste at 2.01 billion tonnes in 2016, projected to 3.40 billion tonnes by 2050 [7]. UNITAR/ITU report 62 million tonnes of e-waste in 2022, with 22.3% documented as formally collected/recycled [8]. UNITAR also reports high embedded material value in e-waste and substantial avoided impacts from formal recycling [9].

Inference from these anchors: for e-waste, the highest-leverage sorting point is typically the collection gate and first formal treatment node ($V^{\text{col}}/V^{\text{sort}}$), because value is concentrated in specific fractions and can be lost rapidly if streams are mixed before depollution/component separation.

B. EU cycle snapshots

Eurostat reports 511 kg municipal waste per capita in 2023 and a 48.0% recycling rate [10]. Packaging waste reached 83.4 million tonnes in 2022; plastic packaging recycling was 41% (14.7 kg recycled per capita out of 36.1 kg generated) [11].

TABLE I
ANCHOR-TO-NODE INTERPRETATION (INFERENCE FROM CITED STATISTICS).

Stream	Quality-loss mechanism if	Candidate high-leverage node v^*	Typical d^*
E-waste	Valuable/hazardous fractions diluted in mixed scrap	Collection gate + first treatment	Medium
Plastic packaging	Polymer/additive mixing lowers secondary-material quality	Source separation + primary sorting	Medium
Corrugated paper	Main risk is contamination, not missing teardown depth	Commercial source segregation	Low
C&D debris	Mineral fractions lose quality when mixed with contaminants	On-site demolition sorting before hauling	Low–Med.

Treated waste in the EU in 2022 split roughly 61.4% recovery and 38.6% disposal [12].

Inference from these anchors: plastics are the main quality bottleneck, so the critical node is usually early material separation (source segregation plus primary sorting), not deep late-stage disassembly. Here, increasing purity by polymer/material class often dominates increasing teardown depth.

C. U.S. cycle snapshots

EPA reports U.S. municipal solid waste at 292.4 million tons in 2018, with 32.1% recycled/composted and about 146.2 million tons landfilled [13]. For paper/paperboard, recycling reached 68.2%, and corrugated boxes 96.5% [14]. EPA also estimates 600 million tons of construction and demolition debris in 2018, with about 455 million tons directed to next use and about 145 million tons landfilled [15].

Inference from these anchors: where a stream is already clean and homogeneous (e.g., corrugated), optimal depth is low and early separation is enough. For C&D, high-volume recovery depends on on-site separation before debris mixing, again pointing to early-node sorting rather than downstream deep disassembly.

D. Norwegian policy-flow example

Mattson et al. show, for Norway, how policy and infrastructure can lock residual waste into incineration-oriented pathways, weakening circular-economy outcomes [16]. In a related flow study for Norwegian WEEE, they estimate materially lower effective recycling when accounting for system leakages than suggested by treatment-only reporting [17]. In model terms, these are direct examples where node definitions and boundary choices change estimated $x_{ij}^{m,q,t}$ and therefore change the measured value of sorting.

VI. INTERPRETATION: THE MAIN PATTERN

Across these economies, the same logic holds:

- 1) optimal sorting location is usually the earliest node that prevents irreversible quality loss (Table I);

- 2) disassembly depth should track stream structure: deep for heterogeneous high-value products (e-waste), shallow for already homogeneous streams (corrugated, mineral C&D fractions);
- 3) stream-specific quality thresholds dominate gross effort: additional labor has value only if it moves flow into a better downstream pathway;
- 4) where downstream offtake is mature, Δ_v rises sharply; where markets are thin or locked into sink pathways, even high sorting effort can yield low Δ_v ;
- 5) disassembly has an interior optimum d_v^* (Fig. 3), not “maximum depth always.”

This is the actionable “aha”: optimize where separation changes pathway quality enough to switch material from low-value sink routes to higher-value recovery routes.

VII. CONCLUSION

Starting from an abstract flow-network class and then binding symbols to real statistics is a workable way to study the value of sorting. The approach is rigorous enough for optimization, yet concrete enough for policy and operations. For a specific product chain (e.g., white goods), the next step is to estimate $\Delta_v(d_v)$ and costs node-by-node, then rank interventions by Δ_v .

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This document was produced as a curiosity-driven synthesis, combining existing literature and public data.

ABOUT THIS DOCUMENT

The contribution is integrative: connect established theory to current empirical flows; do not introduce new edges beyond what existing literature and data already support.

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